WEEK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS
TRAVEL BOOKING MANAGEMENT SYSTEM

Aim

To analyze the relationships between tables in the Travel Booking Management System using SQL joins and generate meaningful reports about customers, travel packages, bookings, payments, and customer reviews.

Introduction

The Travel Booking Management System stores information about customers, travel packages, bookings, payments, and reviews in separate tables. These tables are related through common attributes such as CUSTOMER_ID, BOOKING_ID, and PACKAGE_ID.

SQL joins are used to retrieve related information from multiple tables. They help identify customer bookings, display booking details, analyze payment status, review travel packages, and prepare business reports.

Implement inner join

Concept: An INNER JOIN returns only the records that have matching values in both joined tables.

Task 1: Display customer details along with their bookings. The CUSTOMER and BOOKING tables are joined using CUSTOMER_ID.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,b.booking_id,b.booking_date,b.total_amount,b.booking_status from customer c inner join booking b on c.cus
tomer_id=b.customer_id;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT BOOKING_STATUS
--------- ------------ --------------------
C001                 Arun                 Kumar                B1
10-SEP-26        24000 Confirmed

C002                 Priya                Sharma               B2
11-SEP-26        36000 Confirmed

C003                 Rahul                Raj                  B3
12-SEP-26        55000 Processing


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT BOOKING_STATUS
--------- ------------ --------------------
C004                 Divya                Suresh               B4
13-SEP-26        65000 Confirmed

C005                 Karthik              Mohan                B5
14-SEP-26        25000 Completed

C006                 Sneha                Ravi                 B6
15-SEP-26        30000 Processing


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT BOOKING_STATUS
--------- ------------ --------------------
C007                 Vishnu               Kumar                B7
16-SEP-26        45000 Completed

C008                 Ananya               Krishnan             B8
17-SEP-26        55000 Confirmed


8 rows selected.
```

Task 2: Display booking details with payment information. The BOOKING and PAYMENT tables are joined using BOOKING_ID.

Query:

```
SQL> select b.booking_id,b.customer_id,b.booking_date,b.total_amount,p.payme
nt_id,p.payment_mode,p.payment_status,p.transaction_amount from booking b in
ner join payment p on b.booking_id = p.booking_id;

BOOKING_ID CUSTOMER_ID          BOOKING_D TOTAL_AMOUNT PAYMENT_ID
---------- -------------------- --------- ------------ ----------
PAYMENT_MODE         PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- -------------------- ------------------
B1         C001                 10-SEP-26        24000 P1
UPI                  Success                          24000

B2         C002                 11-SEP-26        36000 P2
Card                 Success                          36000

B3         C003                 12-SEP-26        55000 P3
Net Banking          Pending                          55000


BOOKING_ID CUSTOMER_ID          BOOKING_D TOTAL_AMOUNT PAYMENT_ID
---------- -------------------- --------- ------------ ----------
PAYMENT_MODE         PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- -------------------- ------------------
B4         C004                 13-SEP-26        65000 P4
UPI                  Success                          65000

B5         C005                 14-SEP-26        25000 P5
Card                 Success                          25000

B6         C006                 15-SEP-26        30000 P6
Cash                 Pending                          30000


BOOKING_ID CUSTOMER_ID          BOOKING_D TOTAL_AMOUNT PAYMENT_ID
---------- -------------------- --------- ------------ ----------
PAYMENT_MODE         PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- -------------------- ------------------
B7         C007                 16-SEP-26        45000 P7
UPI                  Success                          45000

B8         C008                 17-SEP-26        55000 P8
UPI                  Failed                           55000


8 rows selected.
```

Task 3: Display customers who have submitted package reviews. The CUSTOMER and REVIEW tables are joined using CUSTOMER_ID.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,r.package_id,r.rating,r.r
eview_comments from customer c inner join review r on c.customer_id=r.custom
er_id;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------
----
C001                 Arun                 Kumar                       101
         5
Excellent travel experience

C002                 Priya                Sharma                      103
         4
Good package and service

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------
----

C003                 Rahul                Raj                         104
         3
Average experience

C004                 Divya                Suresh                      105
         5

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------
----
Wonderful trip and beautiful places

C005                 Karthik              Mohan                       107
         2
Poor accommodation
```

Implement LEFT JOIN

Concept: A LEFT JOIN returns all records from the left table and matching records from the right table. If no matching record exists, the right-side columns contain NULL values.

Task 1: Display all customers, including customers who have not placed bookings.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,b.booking_id,b.booking_da
te,b.total_amount from customer c left join booking b on c.customer_id=b.cus
tomer_id;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C001                 Arun                 Kumar                B1
10-SEP-26        24000

C002                 Priya                Sharma               B2
11-SEP-26        36000

C003                 Rahul                Raj                  B3
12-SEP-26        55000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C004                 Divya                Suresh               B4
13-SEP-26        65000

C005                 Karthik              Mohan                B5
14-SEP-26        25000

C006                 Sneha                Ravi                 B6
15-SEP-26        30000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C007                 Vishnu               Kumar                B7
16-SEP-26        45000

C008                 Ananya               Krishnan             B8
17-SEP-26        55000

C009                 Naveen               Raj
```

```
CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C010                 Meena                Devi



10 rows selected.
```

Task 2: Find customers without any bookings.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name from customer c left join
 booking b on c.customer_id=b.customer_id where b.booking_id is null;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
C009                 Naveen               Raj
C010                 Meena                Devi
```

Task 3: Display all travel packages, including packages without reviews.

Query:

```
SQL> select t.package_id,t.package_name,t.price,r.review_id,r.rating,r.revie
w_comments from travelpackage t left join review r on t.package_id=r.package
_id;

PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------
----
       101
Ooty Holiday Package
     13000 R1                   5
Excellent travel experience


PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------
----
       103
Dubai Holiday Package
     55000 R2                   4
Good package and service
```

```
PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       104
Singapore Tour Package
     65000 R3                   3
Average experience


PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       105
Manali Adventure Trip
     25000 R4                   5
Wonderful trip and beautiful places


PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       107
Goa Honeymoon Package
     30000 R5                   2
Poor accommodation
```

```
PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       111
Tirupati Pilgrimage Tour
      9000



PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       112
Varanasi Spiritual Tour
     14000



PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       109
Mysore Family Package
     15000
```

```
PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       110
Rajasthan Family Tour
     28000



PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       106
Rishikesh Adventure Tour
     22000



PACKAGE_ID
----------
PACKAGE_NAME
--------------------------------------------------------------------------------
----
     PRICE REVIEW_ID      RATING
---------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
----
       108
Kashmir Honeymoon Package
     45000
```

Implement RIGHT JOIN

Concept: A RIGHT JOIN returns all records from the right table and matching records from the left table. If no match exists, the left-side columns contain NULL values.

Task 1: Display all bookings with customer information.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,b.booking_id,b.booking_d
te,b.total_amount from customer c right join booking b on c.customer_id=b.c
stomer_id;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C001                 Arun                 Kumar                B1
10-SEP-26        24000

C002                 Priya                Sharma               B2
11-SEP-26        36000

C003                 Rahul                Raj                  B3
12-SEP-26        55000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C004                 Divya                Suresh               B4
13-SEP-26        65000

C005                 Karthik              Mohan                B5
14-SEP-26        25000

C006                 Sneha                Ravi                 B6
15-SEP-26        30000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C007                 Vishnu               Kumar                B7
16-SEP-26        45000

C008                 Ananya               Krishnan             B8
17-SEP-26        55000


8 rows selected.
```

Task 2: Find bookings where customer details are missing.

Query:

```
SQL> select b.booking_id,b.customer_id,b.booking_date from customer c right
join booking b on c.customer_id=b.customer_id where c.customer_id is null;

no rows selected
```

Task 3: Display all payment records with booking details.

Query:

```
SQL> select b.booking_id,b.customer_id,b.booking_date,p.payment_id,p.payment
_mode,p.payment_status,p.transaction_amount from booking b right join paymen
t p on b.booking_id=p.booking_id;

BOOKING_ID CUSTOMER_ID          BOOKING_D PAYMENT_ID PAYMENT_MODE
---------- -------------------- --------- ---------- --------------------
PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- ------------------
B1         C001                 10-SEP-26 P1         UPI
Success                           24000

B2         C002                 11-SEP-26 P2         Card
Success                           36000

B3         C003                 12-SEP-26 P3         Net Banking
Pending                           55000


BOOKING_ID CUSTOMER_ID          BOOKING_D PAYMENT_ID PAYMENT_MODE
---------- -------------------- --------- ---------- --------------------
PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- ------------------
B4         C004                 13-SEP-26 P4         UPI
Success                           65000

B5         C005                 14-SEP-26 P5         Card
Success                           25000

B6         C006                 15-SEP-26 P6         Cash
Pending                           30000


BOOKING_ID CUSTOMER_ID          BOOKING_D PAYMENT_ID PAYMENT_MODE
---------- -------------------- --------- ---------- --------------------
PAYMENT_STATUS       TRANSACTION_AMOUNT
-------------------- ------------------
B7         C007                 16-SEP-26 P7         UPI
Success                           45000

B8         C008                 17-SEP-26 P8         UPI
Failed                            55000


8 rows selected.
```

Retrieve Complete Booking Details

A complete booking report can be generated by joining the CUSTOMER, BOOKING, and PAYMENT tables. It contains customer name, booking ID, booking date, total booking amount, payment mode, and payment status.

Query:

```
SQL> select c.first_name,c.last_name,b.booking_id,b.booking_date,b.total_amo
unt,p.payment_mode,p.payment_status from customer c inner join booking b on
c.customer_id=b.customer_id left join payment p on b.booking_id=p.booking_id
;

FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D TOTAL_AMOUNT
-------------------- -------------------- ---------- --------- ------------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
Arun                 Kumar                B1         10-SEP-26        24000
UPI                  Success

Priya                Sharma               B2         11-SEP-26        36000
Card                 Success

Rahul                Raj                  B3         12-SEP-26        55000
Net Banking          Pending


FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D TOTAL_AMOUNT
-------------------- -------------------- ---------- --------- ------------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
Divya                Suresh               B4         13-SEP-26        65000
UPI                  Success

Karthik              Mohan                B5         14-SEP-26        25000
Card                 Success

Sneha                Ravi                 B6         15-SEP-26        30000
Cash                 Pending


FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D TOTAL_AMOUNT
-------------------- -------------------- ---------- --------- ------------
PAYMENT_MODE         PAYMENT_STATUS
-------------------- --------------------
Vishnu               Kumar                B7         16-SEP-26        45000
UPI                  Success

Ananya               Krishnan             B8         17-SEP-26        55000
UPI                  Failed


8 rows selected.
```

Display Customer Booking History

Task 1: Display all bookings made by customers.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,b.booking_id,b.booking_da
te,b.total_amount from customer c inner join booking b on c.customer_id=b.cu
stomer_id order by c.customer_id,b.booking_date desc;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C001                 Arun                 Kumar                B1
10-SEP-26        24000

C002                 Priya                Sharma               B2
11-SEP-26        36000

C003                 Rahul                Raj                  B3
12-SEP-26        55000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C004                 Divya                Suresh               B4
13-SEP-26        65000

C005                 Karthik              Mohan                B5
14-SEP-26        25000

C006                 Sneha                Ravi                 B6
15-SEP-26        30000


CUSTOMER_ID          FIRST_NAME           LAST_NAME            BOOKING_ID
-------------------- -------------------- -------------------- ----------
BOOKING_D TOTAL_AMOUNT
--------- ------------
C007                 Vishnu               Kumar                B7
16-SEP-26        45000

C008                 Ananya               Krishnan             B8
17-SEP-26        55000


8 rows selected.
```

Task 2: Display the total amount spent by each customer. LEFT JOIN and SUM include customers without bookings, with their spending displayed as zero.

Query:

```
SQL> select c.customer_id,c.first_name,c.last_name,nvl(sum(b.total_amount),0
) as total_amount_spent from customer c left join booking b on c.customer_id
=b.customer_id group by c.customer_id,c.first_name,c.last_name;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_AMOUNT_SPENT
------------------
C005                 Karthik              Mohan
             25000

C008                 Ananya               Krishnan
             55000

C001                 Arun                 Kumar
             24000


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_AMOUNT_SPENT
------------------
C004                 Divya                Suresh
             65000

C002                 Priya                Sharma
             36000

C003                 Rahul                Raj
             55000


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_AMOUNT_SPENT
------------------
C006                 Sneha                Ravi
             30000

C007                 Vishnu               Kumar
             45000

C010                 Meena                Devi
                 0

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_AMOUNT_SPENT
------------------
C009                 Naveen               Raj
                 0


10 rows selected.
```

Task 3: Display the number of bookings placed by each customer.

Query:

```
SQL>  select c.customer_id,c.first_name,c.last_name,count(b.booking_id) as t
otal_bookings from customer c left join booking b on c.customer_id=b.custome
r_id group by c.customer_id,c.first_name,c.last_name;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            TOTAL_BOOKING
S
-------------------- -------------------- -------------------- -------------
-
C005                 Karthik              Mohan
1
C008                 Ananya               Krishnan
1
C001                 Arun                 Kumar
1
C004                 Divya                Suresh
1
C002                 Priya                Sharma
1
C003                 Rahul                Raj
1
C006                 Sneha                Ravi
1
C007                 Vishnu               Kumar
1
C010                 Meena                Devi
0
C009                 Naveen               Raj
0

10 rows selected.
```

Task 4: Display the latest booking details of customers.

Query:

```
SQL> SELECT B.CUSTOMER_ID, B.BOOKING_ID, B.BOOKING_DATE, B.TOTAL_AMOUNT
  2  FROM BOOKING B
  3  WHERE B.BOOKING_DATE = (
  4      SELECT MAX(B2.BOOKING_DATE)
  5      FROM BOOKING B2
  6      WHERE B2.CUSTOMER_ID = B.CUSTOMER_ID
  7  );

CUSTOMER_ID          BOOKING_ID BOOKING_D TOTAL_AMOUNT
-------------------- ---------- --------- ------------
C001                 B1         10-SEP-26        24000
C002                 B2         11-SEP-26        36000
C003                 B3         12-SEP-26        55000
C004                 B4         13-SEP-26        65000
C005                 B5         14-SEP-26        25000
C006                 B6         15-SEP-26        30000
C007                 B7         16-SEP-26        45000
C008                 B8         17-SEP-26        55000

8 rows selected.
```

Generate Multi-Table Business Reports

Report 1: Customer Booking Report

The CUSTOMER and BOOKING tables are joined to display customer name, booking ID, booking date, and booking status. This report helps track customer bookings.

Query:

```
SQL> SELECT C.FIRST_NAME, C.LAST_NAME, B.BOOKING_ID,
  2         B.BOOKING_DATE, B.BOOKING_STATUS
  3  FROM CUSTOMER C
  4  INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID;

FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D
-------------------- -------------------- ---------- ---------
BOOKING_STATUS
--------------------
Arun                 Kumar                B1         10-SEP-26
Confirmed

Priya                Sharma               B2         11-SEP-26
Confirmed

Rahul                Raj                  B3         12-SEP-26
Processing


FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D
-------------------- -------------------- ---------- ---------
BOOKING_STATUS
--------------------
Divya                Suresh               B4         13-SEP-26
Confirmed

Karthik              Mohan                B5         14-SEP-26
Completed

Sneha                Ravi                 B6         15-SEP-26
Processing


FIRST_NAME           LAST_NAME            BOOKING_ID BOOKING_D
-------------------- -------------------- ---------- ---------
BOOKING_STATUS
--------------------
Vishnu               Kumar                B7         16-SEP-26
Completed

Ananya               Krishnan             B8         17-SEP-26
Confirmed


8 rows selected.
```

Report 2: Booking Amount Report

The booking amount report calculates the total booking amount for each customer using an aggregate function.

Query:

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME,
  2         SUM(B.TOTAL_AMOUNT) AS TOTAL_BOOKING_AMOUNT
  3  FROM CUSTOMER C
  4  INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_BOOKING_AMOUNT
--------------------
C005                 Karthik              Mohan
               25000

C008                 Ananya               Krishnan
               55000

C001                 Arun                 Kumar
               24000


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_BOOKING_AMOUNT
--------------------
C004                 Divya                Suresh
               65000

C002                 Priya                Sharma
               36000

C003                 Rahul                Raj
               55000


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
TOTAL_BOOKING_AMOUNT
--------------------
C006                 Sneha                Ravi
               30000

C007                 Vishnu               Kumar
               45000


8 rows selected.
```

Report 3: Payment Analysis Report

The PAYMENT table is grouped by payment mode. COUNT calculates the number of transactions and conditional SUM counts successful transactions. Replace SUCCESSFUL with the exact successful status stored in your table if different.

Query:

```
SQL> SELECT PAYMENT_MODE, COUNT(*) AS TOTAL_TRANSACTIONS,
  2         SUM(CASE WHEN UPPER(PAYMENT_STATUS) = 'SUCCESSFUL'
  3                  THEN 1 ELSE 0 END) AS SUCCESSFUL_PAYMENTS
  4  FROM PAYMENT
  5  GROUP BY PAYMENT_MODE;

PAYMENT_MODE         TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ------------------ ------------------
Net Banking                           1                  0
Card                                  2                  0
Cash                                  1                  0
UPI                                   4                  0
```

Report 4: Customer Booking Analysis

a. Customers with the highest total booking amount

Query:

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME,
  2         SUM(B.TOTAL_AMOUNT) AS TOTAL_SPENT
  3  FROM CUSTOMER C
  4  INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  6  HAVING SUM(B.TOTAL_AMOUNT) = (
  7      SELECT MAX(CUSTOMER_TOTAL)
  8      FROM (SELECT SUM(TOTAL_AMOUNT) AS CUSTOMER_TOTAL
  9            FROM BOOKING GROUP BY CUSTOMER_ID)
 10  );

CUSTOMER_ID          FIRST_NAME           LAST_NAME            TOTAL_SPENT
-------------------- -------------------- -------------------- -----------
C004                 Divya                Suresh                     65000
```

b. Customers with the maximum number of bookings

Query:

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME,
  2         COUNT(B.BOOKING_ID) AS TOTAL_BOOKINGS
  3  FROM CUSTOMER C
  4  INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  6  HAVING COUNT(B.BOOKING_ID) = (
  7      SELECT MAX(BOOKING_COUNT)
  8      FROM (SELECT COUNT(*) AS BOOKING_COUNT
  9            FROM BOOKING GROUP BY CUSTOMER_ID)
 10  );

CUSTOMER_ID          FIRST_NAME           LAST_NAME            TOTAL_BOOKING
S
-------------------- -------------------- -------------------- -------------
-
C005                 Karthik              Mohan
1
C008                 Ananya               Krishnan
1
C001                 Arun                 Kumar
1
C004                 Divya                Suresh
1
C002                 Priya                Sharma
1
C003                 Rahul                Raj
1
C006                 Sneha                Ravi
1
C007                 Vishnu               Kumar
1

8 rows selected.
```

c. Customers with the highest spending

Query:

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME,
  2         SUM(B.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM CUSTOMER C
  4  INNER JOIN BOOKING B ON C.CUSTOMER_ID = B.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
  6  HAVING SUM(B.TOTAL_AMOUNT) = (
  7      SELECT MAX(CUSTOMER_TOTAL)
  8      FROM (SELECT SUM(TOTAL_AMOUNT) AS CUSTOMER_TOTAL
  9            FROM BOOKING GROUP BY CUSTOMER_ID)
 10  );

CUSTOMER_ID          FIRST_NAME           LAST_NAME            TOTAL_SPENDIN
G
-------------------- -------------------- -------------------- -------------
-
C004                 Divya                Suresh                       6500
0
```

Result

The Week 8 database relationship analysis was carried out using INNER JOIN, LEFT JOIN, and RIGHT JOIN operations on the Travel Booking Management System. Customer booking history, complete booking details, booking amount information, payment analysis, and customer booking reports were generated by combining related tables.

The exercise demonstrated how SQL joins and aggregate functions can be used to retrieve meaningful information from a relational database and support business analysis. The query results depend on the records available in the related tables.

Conclusion

This exercise provided practical experience in analyzing relationships between tables in the Travel Booking Management System using SQL joins. The implementation demonstrated INNER JOIN, LEFT JOIN, RIGHT JOIN, aggregate functions, subqueries, and multi-table business reporting.

The exercise also improved the understanding of how related tables can be combined to retrieve customer booking history, payment details, and useful business information.